

Statistical Data Analysis

Lecture 8: Nonparametric tests 1

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Periods 4 & 5

⏮ Chapter 5: The bootstrap

📍 Chapter 6: Nonparametric methods

- Section 6.1 – The one-sample problem
- Section 6.2 – Asymptotic efficiency
- ▶▶ Section 6.3 – Two-sample problems
- ▶▶ Section 6.4 – Tests for correlation

Nonparametric Tests

Nonparametric Tests – Recall

- › Recall concepts
 - › Actual level (size) of test: (supremum of) probability of type I error
 - › A test has significance level α if its size is $\leq \alpha$
 - › Having the actual level larger than the pre-set α is undesirable
 - › Power of a test: probability of correctly rejecting the null

One-Sample t -Test: Parametric Location Test

- › The one-sample t -test is used to test hypotheses about the population mean
 - › Assumption $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F = N(\mu, \sigma^2)$, with $\sigma > 0$
 - › Null hypothesis $H_0 : \mu = \mu_0$
 - › Test statistic $T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$.
 - › Null distribution: under H_0 , $T \sim t_{n-1}$
 - › Critical region: Reject if $|T| > t_{n-1, \alpha/2}$

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 - › Null distribution: under H_0 , $T \sim t_{n-1}$
 - › Critical region: Reject if $|T| > t_{n-1, \alpha/2}$
- › Problem: if F not normal $\rightsquigarrow$ t approximation flawed
 - › Actual level of t -test in this case?
 - › If $T \sim t_{n-1}$, $P(|T| > t_{n-1, \alpha/2}) \neq \alpha$
 - › Actual power of t -test in this case?

Nonparametric Tests – Idea

- **Problem:** using parametric tests when distributional assumptions **do not** hold $\rightsquigarrow$ **actual** test level and power **deviate** from theoretical ones
- **Remedy:** nonparametric (i.e., **distribution-free**) tests
 - No parametric assumptions about underlying distribution F
 - **Actual** level is α for a broad class of distributions
 - Test statistics **nonparametric** under the null $\rightsquigarrow$ null distribution does not depend on F
 - Greater **power** than parametric tests when their distributional assumption do not hold
 - **Caveat:** parametric tests have greater power when their distributional assumptions hold

t-Test when $F \neq N$ – Illustration

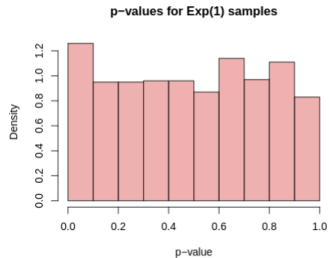
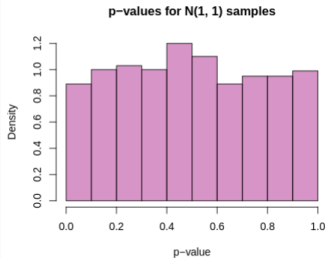
```
# Want to test  $H_0: \mu = 1$  ( $\mu_{\text{zero}}$  is 1), use significance level  $\alpha = 5\%$ 
n <- 20
reps <- 10^3
pval_norm <- numeric(reps); pval_exp <- numeric(reps)
set.seed(123)
for (i in 1:reps){
  # Simulate data from normal with mean 1 and test, extract p_value
  pval_norm[i] <- t.test(rnorm(n, mean = 1, sd = 1), mu = 1)$p.value
  # Simulate data from exponential with mean 1 and test, extract p_value
  # Recall that for exponential distribution, mean = 1/rate
  pval_exp[i] <- t.test(rexp(n, rate = 1), mu = 1)$p.value
}

# Calculate frequency of type I error (i.e., mistakenly rejecting  $H_0$  when true)
paste0("Type I error frequency for N(1, 1) samples: ", sum(pval_norm < 0.05)/reps)

## [1] "Type I error frequency for N(1, 1) samples: 0.044"

paste0("Type I error frequency for Exp(1) samples: ", sum(pval_exp < 0.05)/reps)

## [1] "Type I error frequency for Exp(1) samples: 0.08"
```



Background

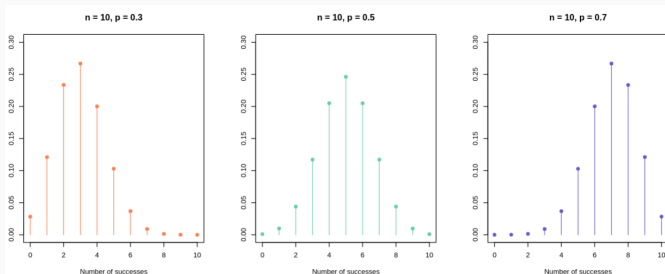
OG Statistical Hypothesis Testing

- In 1710, John Arbuthnot examined London christening records
 - In each of the 82 years from 1629 to 1710, more males than females were born
 - Assuming that the probability of a year with male births exceeding female ones is $1/2$, the probability of the above is $\underbrace{0.5 \times \dots \times 0.5}_{82 \text{ times}} = 2^{-82}$
- Arbuthnot's conclusion: such a small number $\rightsquigarrow$ **not by chance**
 - ☹ His causal reasoning: nature brings forth more males than females as *"the external accidents to which males are subject (who must seek their food with danger) make a great havock of them"*
 - You can read the original paper [here](#) (caveat: recall it is from 1710...)
- Regardless of the author's religious/philosophical interpretations, this is credited as the **first use** of statistical hypothesis testing
 - ❓ Can you identify the null hypothesis?
 - ❓ Can you identify the test statistic and its null distribution?

Binomial Distribution and Binomial Test

- › A Bernoulli trial has two possible outcomes: ‘success’ and ‘failure’
- › If the probability of success is p , the number of successes in n independent Bernoulli trials follows a binomial distribution
- › If $X \sim B(n, p)$, the probability of obtaining exactly k successes is

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} = \frac{n!}{k!(n-k)!} p^k (1 - p)^{n-k}$$



- › **Binomial test** idea: under $H_0 : p = p_0$, one expects np_0 successes
 - › Example: if p is the probability of a ‘male’ year, test $H_0 : p = 1/2$

Binomial Test for Arbuthnot's Data

```
### Arbuthnot's data: nr. of male and female births in London from 1629 to 1710
head(openintro::arbuthnot, n = 3)
```

```
## # A tibble: 3 x 3
##   year boys girls
##   <int> <int> <int>
## 1  1629  5218  4683
## 2  1630  4858  4457
## 3  1631  4422  4102
```

```
n <- nrow(openintro::arbuthnot) # 82 years
# Replicate his reasoning: count 'successes' ('male' years)
k <- sum(openintro::arbuthnot$boys > openintro::arbuthnot$girls) # Also 82
# Test
binom.test(x = k, n = n, p = 0.5, alternative = "two.sided")
```

```
##
## Exact binomial test
##
## data: k and n
## number of successes = 82, number of trials = 82, p-value < 2.2e-16
## alternative hypothesis: true probability of success is not equal to 0.5
## 95 percent confidence interval:
##  0.9560105 1.0000000
## sample estimates:
## probability of success
##                                1
```

And Nowadays?

```
### Dataset present: nr. of male and female births in the USA from 1940 to 2002
head(openintro::present, n = 3)
```

```
## # A tibble: 3 x 3
##   year    boys    girls
##   <dbl>  <dbl>  <dbl>
## 1  1940 1211684 1148715
## 2  1941 1289734 1223693
## 3  1942 1444365 1364631
```

```
n <- nrow(openintro::present) # 63 years
# Replicate his reasoning: count 'successes' ('male' years)
k <- sum(openintro::present$boys > openintro::present$girls) # Also 63
# Test
binom.test(x = k, n = n, p = 0.5, alternative = "two.sided")
```

```
##
## Exact binomial test
##
## data: k and n
## number of successes = 63, number of trials = 63, p-value < 2.2e-16
## alternative hypothesis: true probability of success is not equal to 0.5
## 95 percent confidence interval:
##  0.9431276 1.0000000
## sample estimates:
## probability of success
##                                1
```

Nonparametric Location Tests

Sign Test (for Median)

- › Sign test $\rightsquigarrow$ special case of binomial test where $p_0 = 1/2$
- › Can be used to test hypotheses about the median m
 - › Assume there is **unique** m such that $P(X < m) = P(X > m) = 1/2$
 - › Test $H_0 : m = m_0$
 - › **Remark:** binomial test is for **dichotomous** (binary) categorical data
 - › **Idea:** under the null, $\approx 1/2$ of the data should be **below** m_0 and $\approx 1/2$ of the data should be **above** m_0
 - › **Caveat:** if there are any observations equal to m_0 , they need to be removed from the sample ahead of testing
- › E.g., is the median exam grade significantly different from $m_0 = 7$?

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Sample	7.6	7.5	6.7	6.3	7.2	7.1	5.6	7.8	8.3	8.9
'Sign'	+	+	-	-	+	+	-	+	+	+

Sign Test (for Median)

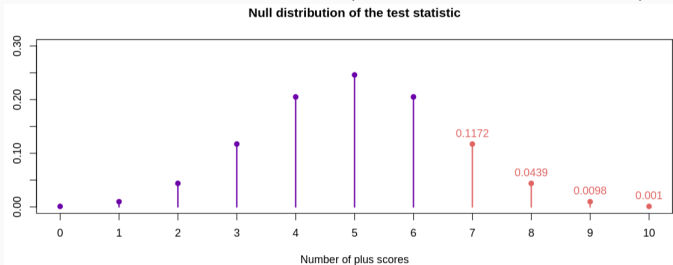
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- › The **test statistic** $T = \#(X_i > m_0)$ is the number of **plus scores**
 - › Null distribution: $T \stackrel{H_0}{\sim} B(n, 0.5)$
 - › It only depends on $n \rightsquigarrow$ sign test for median is **nonparametric**

Sign Test – Illustration

- > In the median grade example, $n = 10$ and we observe 7 plus scores
 - > p -value of two-sided test is ca. $2 \cdot (0.1172 + 0.0439 + 0.0098 + 0.001) = 0.3438$



- > In **R**, we can use `binom.test` as follows:

```
binom.test(x = 7, n = 10, p = 0.5, alternative = "two.sided")

##
## Exact binomial test
##
## data: 7 and 10
## number of successes = 7, number of trials = 10, p-value = 0.3438
## alternative hypothesis: true probability of success is not equal to 0.5
## 95 percent confidence interval:
##  0.3475471 0.9332605
## sample estimates:
## probability of success
##                0.7
```

(Wilcoxon) Signed Rank Test

- › (Wilcoxon) Signed Rank Test makes use of additional information
 - › Uses not only **sign**, but also **absolute value** of deviations
 - › **Assumption**: symmetric distribution $\rightsquigarrow$ test hypotheses about point of symmetry, such as $H_0 : m = m_0$ vs. $H_1 : m \neq m_0$
 - › **Caveat**: if there are any observations equal to m_0 , they need to be removed from the sample ahead of testing

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 - › **Caveat**: if there are any observations equal to m_0 , they need to be removed from the sample ahead of testing
- › **Test statistic**: $V = \sum_{i=1}^n R_i \operatorname{sgn}(X_i - m_0)$, where R_i is rank of $|X_i - m_0|$
 - › $\operatorname{sgn}(X_i - m_0)$ being 1 (resp. -1) indicates $m > m_0$ (resp. $m < m_0$)
 - › This indication is stronger when $|X_i - m_0|$ (and hence R_i) is larger
 - › Large values of V indicate that $m > m_0$, and viceversa for small ones

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 - › Large values of V indicate that $m > m_0$, and viceversa for small ones
- › **Null distribution** of V : under H_0 , $V \stackrel{d}{=} \sum_{i=1}^n Q_i \tilde{R}_i$
 - › $Q_i, i = 1, \dots, n$, i.i.d. with $\mathbb{P}(Q_i = -1) = \mathbb{P}(Q_i = 1) = 1/2$
 - › $(\tilde{R}_1, \dots, \tilde{R}_n)$ a random permutation of $(1, \dots, n)$
 - › It only depends on $n \rightsquigarrow$ Wilcoxon test is **nonparametric**
 - › For large n and under H_0 , $V \overset{\text{approx.}}{\sim} N\left(0, \frac{n(n+1)(2n+1)}{6}\right)$

(Wilcoxon) Signed Rank Test – Illustration

- Note: the positive-rank sum and negative-rank sum are equivalent test statistics

- $V_+ = \sum_{i: X_i > m_0} R_i$ and $V_- = \sum_{i: X_i < m_0} R_i$.
- $V = 2V_+ - \frac{1}{2}n(n+1) = \frac{1}{2}n(n+1) - 2V_-$
- Corresponding tests are also nonparametric
- Different null distributions, but same testing decision (same p -value)

- In **R**: `wilcox.test` uses the positive-rank sum test statistic

```
# Recall m_null is 7
wilcox.test(x = grades, mu = m_null, alternative = "two.sided")

##
## Wilcoxon signed rank exact test
##
## data: grades
## V = 37, p-value = 0.375
## alternative hypothesis: true location is not equal to 7
```

Wilcoxon Test vs. Sign Test vs. t -Test – p -values

- › One may wonder how the different tests compare:

```
# Recall m_null is 7
t.test(grades, mu = m_null)$p.value

## [1] 0.3473628

binom.test(plus_scores, n_grades, p = 0.5, alternative = "two.sided")$p.value

## [1] 0.34375

wilcox.test(grades, mu = m_null, alternative = "two.sided")$p.value

## [1] 0.375
```

❓ Not much difference in p -values in this case... Which one should one choose?

- › **Caveat:** recall first section of this lecture
- › Need **correct significance level**, **high power** desirable
- › **Power** of a test: probability of **correctly** rejecting the null
- › Higher **power** for same $n \rightsquigarrow$ **more efficient** test
- › We will now study this concept **asymptotically** ($n \rightarrow \infty$)

Asymptotic Efficiency

Asymptotic Efficiency – Illustration

- › General problem: $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} F$
 - › According to H_0 , $F \in \mathcal{F}_0$ (e.g., class of all distributions with median m_0)
 - › Test power given by the function $\pi(F) = \mathbb{P}_F(H_0 \text{ is rejected})$
 - › Should be small when $F \in \mathcal{F}_0$ and large when $F \in \mathcal{F}_1$
 - › Hard to compare power of two tests for every possible F
 - › Note: here we focus on (location) shift alternatives
 - › Fix $F_0 \in \mathcal{F}_0$, alternatives are obtained by shifting F_0 over a certain distance θ

Asymptotic Efficiency – Illustration

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 - › Fix $F_0 \in \mathcal{F}_0$, alternatives are obtained by shifting F_0 over a certain distance θ
- › Statistical problem: $H_0 : \theta \leq 0$ vs. $H_1 : \theta > 0$
 - › Similar reasoning applies to left-sided and two-sided problems
- › T_n : asymptotically normal test statistic (we consider $n \rightarrow \infty$)
 - › Asymptotic mean: $\mu(\theta)$, asymptotic standard deviation: $\sigma(\theta)$
 - › Thus, $\sqrt{n} \frac{T_n - \mu(\theta)}{\sigma(\theta)} \stackrel{\text{large } n}{\sim} N(0, 1)$
 - › Reject H_0 for large $T_n \rightsquigarrow$ if $\sqrt{n} \frac{T_n - \mu(0)}{\sigma(0)} > \xi_{1-\alpha} := \Phi^{-1}(1 - \alpha)$ (test level: α)
 - › The true value of the parameter is $\theta \rightsquigarrow$ see use of $\mathbb{P}_\theta$

Asymptotic Efficiency – Asymptotics

- › Power of the (asymptotic version of the) test:

$$\begin{aligned}\mathbb{P}_\theta \left(\sqrt{n} \frac{T_n - \mu(0)}{\sigma(0)} > \xi_{1-\alpha} \right) &= \mathbb{P}_\theta \left(\sqrt{n} \frac{T_n - \mu(\theta)}{\sigma(\theta)} > \xi_{1-\alpha} \cdot \frac{\sigma(0)}{\sigma(\theta)} + \sqrt{n} \frac{\mu(0) - \mu(\theta)}{\sigma(\theta)} \right) \\ &\approx 1 - \Phi \left(\xi_{1-\alpha} \cdot \frac{\sigma(0)}{\sigma(\theta)} + \sqrt{n} \frac{\mu(0) - \mu(\theta)}{\sigma(\theta)} \right)\end{aligned}$$

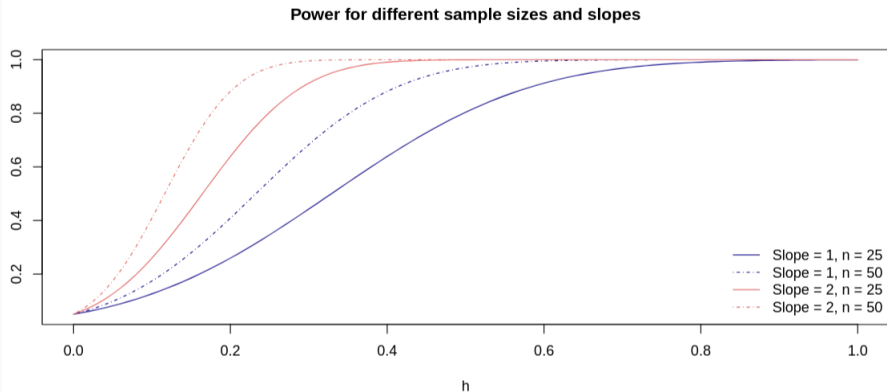
- › The approximation is due to T_n being asymptotically normal
- › Consider, for fixed h , $\theta_n = h/\sqrt{n}$ (goes $\rightarrow 0$, but not too rapidly)
 - › $\frac{\mu(h/\sqrt{n}) - \mu(0)}{h/\sqrt{n}} \xrightarrow{n \rightarrow \infty} \mu'(0)$
 - › $\sigma(h/\sqrt{n}) \xrightarrow{n \rightarrow \infty} \sigma(0)$
- › Thus, asymptotically:

$$\text{power} \stackrel{\text{large } n}{\approx} 1 - \Phi \left(\xi_{1-\alpha} - \frac{\mu'(0)}{\sigma(0)} \cdot h \right)$$

- › We can study the power $\pi_n(\theta_n) = \pi_n(h/\sqrt{n})$ for varying h and large n
- › The quantity $\mu'(0)/\sigma(0)$ is the slope of the test

Slope of a Test

- › Slope of test: $\mu'(0)/\sigma(0) \rightsquigarrow$ when h increases, the power increases faster when slope is larger



- › A smaller test slope can be compensated by a larger sample size

Summary and wrap-up

Summary

- › Nonparametric tests

- › Provide a more reliable p -value when distributional assumptions fail

- › Sign test

- › Tests $H_0 : m = m_0$ for the median m
 - › Computes $\# \{X_i > m_0\}$

- › (Wilcoxon) signed rank tests

- › Expands on the sign test by including the ranks of $|X_i - m_0|$

- › Asymptotic efficiency

- › Gives us an idea of the **power** of a test when the test statistic is asymptotically normal

What now?

> To do for now

- > Start with assignment 5
- > Deadline: 14 April 23:59
- > Good luck!

> Tutorials

- > No tutorial this week (good friday)
- > After that. Only 1 TA present in 4B05, 09:00 – 11:00.